INCH-POUND

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23 February 1994
SUPERSEDING
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(see 6.10)

MILITARY SPECIFICATION

DRY CLEANING UNIT ASSEMBLY, FULLY AUTOMATIC, SYNTHETIC SOLVENT PERCHLORETHYLENE TYPE, (20-35 POUNDS CAPACITY) (SHIPBOARD)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fully automatic dry-cleaning machines of the dry-to-dry, closed solvent recovery type, for using perchlorethylene solvent, for Navy shipboard use.

1.2 Classification. The dry cleaning machine covered by this specification shall be of the following types, classes, and styles, as specified (see 6.2):

Type I - Electro-mechanical automatic formula control unit.
Type II - Microprocessor automatic formula control unit.

Class 1 - Unsectionalized
Class 2 - Sectionalized

Style a - 20-25 pound dry weight
Style b - 30-35 pound dry weight

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 03R42, Department of the Navy, 2531 Jefferson Davis Hwy., Washington, DC 20362-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3510

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-F-320	Fiberboard: Corrugated and Solid, Sheet Stock container Grade) and Cut Shapes.
QQ-N-281	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.

MILITARY

MIL-L-3153	Laundry and Dry Cleaning Machinery and Equipment, Packaging of.
MIL-C-15726	Copper-Nickel Alloy, Sheet, Plate, Strip, Bar Rod, and Wire.
MIL-T-16420	Tube, Copper-Nickel Alloy, Seamless and Welded- (Copper Alloy Numbers 715 and 706).
MIL-L-19140	Lumber and Plywood, Fire-Retardant Treated.
MIL-H-24135/10	Hose, Synthetic Rubber, Wire Reinforced, and End Fittings, Reusable, for Flexible Hose Assemblies (Single or Double Wire Braid, Medium Pressure, Sizes-4 through - 48).
MIL-H-24136/3	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, and End Fittings, Reusable, for Flexible Hose Assemblies (Low Pressure, Sizes - 4 through - 32).

STANDARDS

MILITARY

MIL-STD-108	-	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I Environmental and Type II Internally Excited).
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electro-magnetic Interference.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.

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MIL-STD-740-1	-	Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
DOD-STD-1399, Section 300	-	Interface Standard for Shipboard Systems Section 300A, Electric Power, Alternating Current (Metric).

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted shall be those listed in the issue of DODISS cited in the solicitation. Unless otherwise specified, the issue of documents not listed in DODISS shall be the issues of the documents cited In the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section II Material Specifications, Section VIII - Rules for Construction of Pressure Vessels.

(Applications for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017) .

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 123	-	Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products El (DOD Adopted).
A 167	-	Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip (DOD Adopted).
A 269	-	Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service (DOD Adopted).
A 525	-	Standard Specification for General Requirements for Steel Sheet, Zinc-Coated (Galvanized) by the Hot-Dip Process (Metric) (DOD Adopted).
B 61	-	Standard Specification for Steam or Valve Bronze Castings (DOD Adopted).
B 62	-	Standard Specification for Composition Bronze or Ounce Metal Castings.
B 148	-	Standard Specification for Aluminum-Bronze Sand Castings El (DOD Adopted).
B 150	-	Standard Specification for Aluminum-Bronze Rod, Bar, and Shapes (Metric) (DOD Adopted).
B 169	-	Standard Specification for Aluminum-Bronze Plate, Sheet, Strip, and Rolled Bar (Metric) (DOD Adopted).

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B 584 - Standard Specification for Copper Alloy Sand Castings *for* General Applications (DOD Adopted).
F 1166 - Standard Practice for Human Engineering Design for Marine Systems, Equipment and Facilities (DOD Adopted).

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS, INC. (IEEE)

Standard 45 - Recommended Practice for Electrical Installations on Shipboard

(Applications for copies should be addressed to the Institute of Electrical and Electronics Engineers, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG-1 - Motors and Generators

(Applications for copies should be addressed *to* National Electrical Manufacturing Association, 2101 L Street, N. W., Washington, DC 20037).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AS 604 - Hose Assembly, Tetrafluoroethylene 400 Degree F, 3000 psi Hydraulic, Heavy Braid (ANSI Approved) (DOD Adopted).
AS 620 - High Temperature Hose Assembly, Convoluted Tetrafluoroethylene, for Aircraft (ANSI Approved).

(Applications for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in *or* through libraries or other informational services.)

2.3 order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or KS standards), the text of this document takes precedence. Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article When specified (sea 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.4.

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3.2 Materials. Materials shall be as specified in 3.2.1 through 3.2.13.

3-2.1 Structure Frame base, and strength members shall be constructed of malleable iron, ductile iron, or steel hot-dipped galvanized in accordance with ASTM A 525 for corrosion protection. Gray cast iron or other brittle materials shall not be used for these parts. Except as otherwise specified herein, gray cast iron may be used for other components such as gear reducers and bearing housings, which are not strength members unless manufactured as an Integral part of the frame, and satisfactorily completes the vibration test procedures specified here in. Aluminum castings shall not be used.

3.2.2 shell. The shell shall be made of stainless steel in accordance with ASTM A 167, type 302 or 304, or nickel-copper alloy (not less than 60 percent nickel). Front of the shell shall be of stainless steel metal or ferrous metal with stain-less steel liners on the side exposed to the washing solution. The shell shall be provided with a nickel-copper alloy (not less than 60 percent nickel) or stainless steel door frame.

3.2.2.1 Shell loading door. Door and fittings shall be of the same material as used for the shell construction, The viewing window shall be tempered glass of a quality and thickness to prevent damage under normal operation. The door shall be mounted on corrosion resisting heavy duty hinges.

3.2.3 cylinder. The cylinder and lifting ribs shall be made of stainless steel in accordance with ASTM A 167, type 302 or 304, or nickel-copper alloy (not less than 60 percent nickel). The driving trunnion and spider shall be of stainless steel (see 3.4.2.1.1).

3.2.4 Metal joints. Rivets, bolts, machine screws, nuts, and washers shall be of steel, except where brass or corrosion-resisting metal is fastened, in which case they shall be of brass or corrosion-resisting metal. Where dissimilar metals are fastened, rivets, bolts, machine screws, nuts and washers shall be galvanically compatible with materials being joined. Electrodes or welding rods used for welding shall deposit material similar to the base metal. Nuts subject to shock or vibration shall be locked. Welded steel parts, required to be galvanized, shall be galvanized after completion of welding in accordance with ASTM A 123. Drill point screws are prohibited. Spacing of fasteners shall provide strength, tightness, and rattle-free performance. Fasteners of parts through which servicing access is required shall be provided for convenient access.

3.2.5 Pipes, fittings, valves and connection. Unless Otherwise specified pipes, fittings, valves and connections shall be as specified in 3.2.5.1 through 3.2.5.6.

3.2.5.1 Steam lines. Steam pipes shall be made of copper-nickel (90-10) in accordance with MIL-T-16420. Fittings shall be bronze in accordance with ASTM B 61 or ASTM B 62. Valves shall be cast bronze in accordance with ASTM B 61.

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3.2.5.2 Flexible steam hose assemblies. Flexible steam hose assemblies shall be extruded or convoluted tetrafluoroethylene hose with stainless steel wire braid and end fittings in accordance with SAE AS 604 or AS 620.

3.2.5.3 Saltwater cooling lines. Saltwater pipes shall be made of stainless steel, grade TP 304, in accordance with ASTM A 269 or of copper-nickel (90-10) in accordance with MIL-T-16420. Fittings shall be made of the same material as piping. Valves shall be cast bronze in accordance with ASTM B 61, internal metal parts subject to corrosion or erosion shall be made of nickel-copper conforming to QQ-N-281.

3.2.5.4 Flexible water hose assemblies. Flexible water hose assemblies shall be in accordance with MIL-H-24135/10 or MIL-H-24136/3.

3.2.5.5 Solvent lines. Solvent line piping shall be made of stainless steel, grade TP 304, in accordance with ASTM A 269. Fittings and valves shall be as specified in 3.2.5.3.

3.2.5.6 Air lines. Air line tubing shall be made of copper.

3.2.6 Button trap and water separator . The button trap and water separator shall be made of stainless steel in accordance with ASTM A 167, type 302 or 304 or nickel-copper alloy (not less than 60 percent nickel).

3.2.7 Refrigeration recovery circuit .

3.2.7.1 Condenser unit. The condenser shall be designed for use with sea water cooling at an entering temperature of 85° Fahrenheit. All parts in contact with sea water shall be made of copper-nickel (90-10) conforming to MIL-C-15726, aluminum bronze conforming to ASTM B 169, ASTM B 584 alloy 1B, ASTM B 148 alloy 91), ASTM B 150 alloy 2, or valve bronze conforming to ASTM B 61. The condenser seawater tubes shall be copper-nickel (90-10) in accordance with 3.2.5.3. The water regulating valve body shall be made of valve bronze conforming to ASTM B61. In addition, the internal q etal parts subject to corrosion or erosion shall be made of nickel-copper conforming to QQ-N-281. The strainer body shall be valve bronze conforming-to ASTM B 61, copper-nickel (90-10) or monel. A strainer of No. 20 mesh nickel-copper screen of substantial area shall be provided.

3.2.7.2 Evaporator coil. Evaporator cooling coil assembly shall be made of copper tube with aluminum or copper fins.

3.2.7.3 Heating coil. The heating coil assembly shall be made of copper tube with aluminum or copper fins.

3.2.7.4 Drying air steam heating coil. Drying air steam heating coil shall be made of copper tube with aluminum or copper fins in accordance with ASME Boiler and Pressure Vessel Code, Section II.

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3.2.8 Base storage tank. Base storage tank shall be constructed of heavy gauge stainless steel in accordance with ASTM A 167 or of heavy gauge steel plate with stainless steel liners on the side exposed to perchloroethylene. Base tank cooling coil shall be in accordance with 3.2.5.3.

3.2.9 Recovery housing. Recovery housing shall be made of stainless steel in accordance with ASTM A 167.

3.2.10 Filtration system.

3.2.10.1 Filter housing. Filter housing shall be constructed of stainless steel in accordance with ASTM A 167 or nickel-copper alloy (minimum 60 percent nickel).

3.2.10.2 Filter cartridge. Filters shall be paper with a solid carbon core.

3.2.11 Distillation system.

3.2.11.1 Still tank. Still tank shall be constructed of stainless steel in accordance with ASTM A 167 or of ferrous metal with stainless steel metal liners on interior of tank. The still heating coil shall be in accordance with 3.2.5.1. Tank and coil shall be in accordance with ASME Boiler and Pressure Vessel Code, Section II.

3.2.11.2 Still condenser. Still condenser shell be made of stainless steel in accordance with ASTM A 167. The still condenser coil shall be in accordance with paragraph 3.2.5.3.

3.2.12 Recovered material. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practical without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials , as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specified.

3.2.13 Restricted material. The machine shall be free from asbestos, wood or cadmium parts or coatings in any material form.

3.3 Design. The complete machine and its components shall be new and shall be the current standard product of the manufacturer, modified as necessary to meet the requirements of this specification.

3.3.1 Construction. The machine shall be a self-contained integral unit, complete with combination washer-extractor-dryer, closed circuit refrigeration-heat pump solvent recovery system, still, filter, solvent storage tanks, vapor. containment system, button trap, water separator, pumps, motors, gauges and all

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necessary pipes, valves and fittings. The machine shall be front loading, fully automatic, shell and open pocket cylinder with a rotating action about the horizontal axis. Rotation of the Cylinder shall incorporate reversing action during appropriate parts of the production cycle. The machine shall be designed for using salt water cooling at an entering temperature of 85 degree Fahrenheit and shall operate efficiently with perchlorethylene cleaning solvent. Each machine shall be fitted with features as specified herein.

3.3.2 Automatic controller. Each machine shall be furnished with an automatic formula control unit for controlling the machine through any complete cycle of fill, wash, drain, extract, dry, deodorize, filter and distill (see 3.4.14).

3.3.3 Filter and still. Each machine shall be equipped with a solvent filtration and distillation system of a size and capacity suitable for the intended use.

3.3.4 shell door interlock. The shell door shall be fitted with an electric interlock in accordance with 3.4.13.4.

3.3.5 Automatic valves. Drain valves, solvent inlet valve, steam valves and cooling water valves shall be fully automatic. Valves shall be electrically or pneumatically operated. Thermostatic valves shall be specified.

3.3.6 Capacity. The dry weight capacity rating shall be not greater than 3.3 pounds dry weight per cubic foot (3.3 lb/cu ft).

3.3.7 Size and weight. The machine's overall dimensions shall be not greater than 76 inches in height, 65 inches in width (across front) and 48 inches in depth (front to back) for Style a machines and 84 inches in height, 85 inches in width and 52 inches in depth for Style b machines. The dry weight of the machine" including all components shall be not greater than 2750 pounds for Style a machines and 3700 pounds for Style b machines.

3.3.8 Vibration switch. The nachine shall have an adjustable vibration limit cut-out device in accordance with 3.4.13,6,

3.3.9 Vapor containment Each machine shall be equipped with a vapor containment system that prevents the escape of solvent vapors from the cylinder when removing the cleaned load. The vapor containment system shall be interlocked with the loading door. When the door is opened the air in the cylinder basket shall be drawn through slots surrounding the door and into l carbon adsorption filtration system.

3.3.10 Machine mounting. The machine shall be designed for deck mounting by providing not less than four mounting bolt connections 1/2 inch in nominal diameter. No other attachment points are permitted except for water lines, air lines, steam lines, electrical cabling and drain lines. Base structure of mounting pads shall be secured to an area of the machine to ensure adherence of the machine and it's attachments under ship motion and vibration, Structure penetrated by the mounting bolts shall be made of steel and shall be constructed not to exceed the yield point of it's material when subjected to accelerations of 31 G's vertically and 16 G's horizontally about the center of gravity of the machine.

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3.3.11 Controls, indicator lights and audible signals. Controls, indicator lights and audible signals shall be selected and integrated into the machine In order to meet the applicable human engineering requirements specified in ASTM F 1166 and 3.4.14.9.

3.3.12 Finish and color. The machine shall be finished for corrosion resistance according to the manufacturer's standard commercial practice.

3.3.13 Sectionalization. Class 2 machines shall be constructed to be disassembled, transported, and reassembled within the ship without hot work, such as welding or brazing. Major subassemblies that bolt together shall be preferred. Limiting dimensions for each major subassembly shall be 50 inches high, by 25 inches wide, by 36 inches deep. Subassemblies shall be match- marked to assist in field reassembly once in the designated space. The manufacturer shall provide touch-up paint in the event of chipped or scratched paint during breakdown and assembly of machine. A replacement set of bolts, nuts and other associated fasteners or hardware shall be provided in the case of lost or damaged items during breakdown and assembly.

3.3.14 Human factors engineering. Human engineering criteria, principles, and practices, as specified in ASTM F 1166, shall be used in the configuration of the laundry dry cleaning machine where specified throughout this document. The dry cleaning machine shall be constructed so that the access for operation, maintenance, and repair shall be analyzed in terms of the applicable requirements as specified in ASTM F 1166. Maintenance and operational activities shall permit performance by the 5th percentile female to the 95th percentile male as specified in ASTM F 1166. Access to each control, lubrication fitting, or any other component shall include space for the tools and the test equipment required, as well as any protective clothing likely to be worn by the maintainer. Particular attention shall be given to the ease of access for maintenance and removal of cartridge filters, belts and chains, heating and cooling coils and refrigeration compressor . Skill level and training requirements for operation, maintenance and repair shall be kept at the lowest level consistent with other requirements herein. The analysis of operation, maintenance and repair functions shall Include and reflect considerations of safety.

3.4 Details of units or parts. Units or parts shall be as specified in 3.4.1. through 3.4.14.

3.4.1 Shell The shell shall be solvent tight and of sufficient thickness and rigidity to prevent distortion or objectionable vibration under the normal operations onboard ship. Plating shall be continuous in the lower half of the shell. The shell shall be provided with connections for piping, and shall be reinforced where necessary to properly support attachments. Shell seams shall not pass through drain valve opening. The front of the shell shall be removable to permit replacement of the cylinder if replacement can not be accomplished from rear within clearance requirements for filter replacement.

3.4.1.1 Shell openings. The shell shall have openings for a door, cylinder shaft, solvent fill piping, drain outlet, recovery housing, solvent level indicator and temperature indicator.

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3.4.1.2 Shell door. The shell door shall be in accordance with 3.2.2.1, complete with latching means to ensure solvent tightness when closed. Shall door shall' latch in both the open and closed positions. The use of magnetic securing devices is prohibited. For door interlock see 3.4.13.4.

3.4.1.3 Fill and drain connections. Fill and drain connections shall be of sizes suitable for the purpose intended. The design shall be such that the clean solvent will flow into the shell not less than 3 inches above the maximum solvent level and the dirty solvent can be drained from the lowest point in the bottom of the shell through a normally opened, automatically controlled drain valve.

3.4.2 Cylinder. The cylinder shall be rigid to prevent distortion or failure from stresses created during the extract cycle at the rated capacity. Perforations in the cylinder shall be sized and spaced for efficient washing, extraction and drying. Perforations shall be free from burrs, sharp edges or imperfections that will adversely affect clothing. The back sheet of the cylinder shall be solid and the front sheet of the cylinder shall have a circular opening to match the shell door opening. The front of the cylinder shall align with the shell opening in such a way to prevent laundry from falling between the cylinder and the shell. The cylinder shall be fitted with lifting ribs equally spaced around the cylinder.

3.4.2.1 Cylinder drive. The cylinder shall be driven by an electric two speed motor through V-belts or a combination of gearing and V-belts to provide proper speed for the washing, extracting and drying cycles. The cylinder shall reverse rotation not less than three times per 95 seconds during the washing and drying cycles.

3.4.2.1.1 Cylinder driving trunnion. The cylinder driving trunnion shall be made of hard, strong steel (see 3.2.3) welded to a steel transmitter and spider assembly which shall be securely bolted to the rear cylinder plate. Driving trunnion shall provide safe and durable operation that will withstand ship-induced vibrations. Drive shaft shall be machined for proper fit with bearing sleeves at back of machine. The cylinder drive shaft opening' shall be sized according to the shaft size and fittings. Shaft opening shall be sealed to prevent solvent leakage during machine operation.

3.4.2.1.2 Gearing. The type of gearing and the material of the gears shall be selected to provide strength, durability, resistance to wear, and quietness of operation.

3.4.2.1.3 Bearing. Cylinder trunnion bearings shall be self-aligning, shielded, antifriction bearing type. Bearings shall be properly sized and readily replaceable. Seals shall be provided to make bearings solvent proof.

3.4.2.1.4 V-belts. V-belt drives shall be selected in accordance with the belt manufacturer's recommendations for load rating, including applicable safety factors, to drive the machine under the most severe operating conditions. The V-belts shall perform for a service life of not less than 5,000 hours of operation. V-belts shall be readily adjustable.

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3.4.3 Brake. The machine shall be fitted with a electric or pneumatic operatad brake for rapid decelaration and stopping of the cylinder. The brake shall stop the rotation of a loaded cylinder during any part of the extract cycle within 60 seconds. The temperature rise of the friction surface of the brake lining shall be not greater than 300 degrees Fahrenheit (°F) above the initial temperature of the brake lining as a result of three stops from peak speeds at 2-minute intervals during the extract cycle.

3.4.4 Cooling water piping and connections. Cooling water piping shall be configured to provide a smooth flow of water, i.e. tight U-bends or similar configuration is prohibited. The cooling water piping shall have no internal connections within the base storage tank, still condenser and refrigeration condenser. Pressure regulators and pressure gages shall be installed on the inlet side of each cooling water connection. Coils shall be suitable for a working pressure of not greater than 45 lb/sq in. The machine shall be fitted with threaded supply and return connections of a size suitable for the intended use to the base storage tank, refrigeration condenser and still condenser. The flow of winter shall be automatically controlled by electric or pneumatic valves. Valves shall be easily accessible for maintenance and replacement.

3.4.5 Steam heating units. The machine shall be fitted with threaded supply and return connections of a size suitable for the intended use to the still and the drying air heating coil. The flow of steam to the drying air heating coil shall be automatically controlled by electric thermostatic solenoid valves. A strainer shall be furnished with the steam piping. The still steam heat exchanger shall be a closed loop type; steam shall not come in contact with sol-vent. The air heating coil shall consist of a continuous tube with closely spaced fins for efficient heat transfer. The air heating coil shall be located in a position in the recovery housing to evenly heat and distribute air entering the cylinder and to efficiently vaporize the solvent and remove and dissipate solvent odors. Heating unit assembly shall be constructed as specified in ASME Boiler and Pressure Vessel Code, section VIII. Steam systems shall be hydrostatically tested as specified in 4.4.2.7. During testing, no leaks shall be permitted in the steam system. Pipes, fittings, valves, coils and chambers using steam shall be suitable for 100lb/in sq saturated steam working pressure.

3.4.6 Blower. A two speed blower shall be provided to meet the drying and deodorizing requirements.

3.4.7 Refrigeration and recovery circuit.

3.4.7.1 Condensing unit. The condensing unit shall be designed to operate on the frequency and voltage specified The use of any transformer to obtain the specified frequency and voltage is prohibited.

3.4.7.1.1 Compressor. The compressor shall be hermetically sealed. The refrigeration system shall be fitted with a means of discharging and adding refrigerant to the system. When receivers are used, the compressor shall be fitted with auction and discharge valves. The assembled refrigeration system shall be completely dehydrated and charged with the amount of refrigerant and oil necessary for operation. The compressor motor, up to and including 3/4 horsepower shall operate on a nominal 120-volt, 60 hertz, l-phase power source. Compressor motors

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over 3/4 horsepower shall operate on a nominal 440-volt, 60 hertz, 3-phase power source. Motors shall be protected against damage from overloads and voltage fluctuations when starting or running by means of a thermal overload protection and low voltage protection relay.

3.4.7.1.2 Condenser The condenser shall be water cooled of the shell and coil type, the multipass shell and tube type or the tube in tube type with provisions for cleaning the water circuit without opening the refrigerant circuit. Water cooled condenser shall be equipped with a water regulating valve. The condenser shall be designed for use with sea water at an entering temperature of 85°F and not greater than 110°F condensing temperature. Water velocities in the condenser shall meet the condensing capacity with sea water entering the condenser at 90°F. The condenser shall be a type in which the tubes can be mechanically cleaned and shall be Installed so that the tubes may be cleaned without removing the condenser from the equipment. Material in contact with sea water shall be as specified in 3.2.7.1. The condenser shall be provided with specific devices to purge both the refrigerant and water circuits and for draining the water circuit. A water-regulating valve shall be provided with each condenser The valve shall be of the direct-acting or pilot-controlled type actuated by condenser gas pressure to modulate the flow of water required for the condenser. The valve shall be selected to regulate from shut off to required capacity within an operating gas pressure rise of not greater than 60 pounds per square inch (psi). The valve shall be adjus-table within an operating range of 155 to 225 psi refrigerant gas pressure for R-22. The capacity of the water regulating valve shall be not less than Or equal to the water requirement of the condenser, based on available inlet water pressure of 35 psi and a pressure drop across the valve of not greater than 10 psi. Valve material shall be as specified in 3.2.7.1. The valve shall have flanged connections or union ends. The valve shall be constructed in such a manner as to prevent the possibility of entry of sea water to the refrigerant system in the event of derangement. The strainer basket shall be readily removable for cleaning without disassembly of connecting pipe. Receivers, if used, shall be fitted with a back seating outlet valve, a 1/8 inch National Pipe Threat (NPT) opening, closed with a fusible plug. Combination water cooled receiver/condensers are acceptable.

3.4.7.1.3 Evaporator cooling coil. The evaporator cooling coil shall consist of a continuous tube with closely spaced fins for efficient heat transfer. The evaporator cooling coil shall be located in a position in the recovery housing to efficiently condense solvent vapor rising from the cylinder. A thermostatic expansion valve shall be provided to regulate the flow of refrigerant through the coil .

3.4.7.2 Lint filter. A foam filter or lint bag shall be mounted in the recovery housing to trap lint prior to air passing through the evaporator coil. A hinged access door shall be installed for filter maintenance. The filter shall be readily removable.

3.4.8 Base storage tank. The base storage tank shall be two independent solvent tanks. The capacity of each tank shall be not less than 30 gallons for Style a machines and 50 gallons for Style b machines. The storage tank shall be fitted with button and lint trap, solvent level gauge and clean out plate for

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cleaning out the tank. The storage tanks shall be supplied with a closed loop in-tank cooling coil to maintain optimum solvent temperature. A thermostatic valve shall be installed on the cooling coil to open automatically with the storage tank above 75°F and closes with the storage tank below 60°F.

3.4.9 Still. The still shall have a holding capacity of not less than 30 gallons for Style a machines and 50 gallons for Style b machine. Distillation rate shall not be less than 50 gallons per hour. The still shall be able to be filled directly from the washer shell, filter or base tank. Steam supply shall be con-trolled by an automatic solenoid valve that will be continuously opened during the distillation process. A float assembly shall be installed in the still that will shut down the operation if a preset high level is reached to prevent solvent over-flow into condenser and water separator. A thermostat shall be located on the solvent outlet of the condenser to the water separator to ensure proper water flow to condenser. Steam supply shall be automatically shut off if improper amounts of water are being supplied. The still shall be fitted with a drain and clean out cover to remove residue and oil buildup. An illuminated sight glass shall be pro-vialed to allow monitoring of still operation and residue buildup. A pressure relief valve shall be fitted on the still that will release at pressure of 7 pounds per square inch.

3.4.10 Filter. Filtration canisters shall accommodate four paper/carbon core filter cartridges for Style a machines and eight cartridges for Style b machines. The filter canisters shall recirculate solvent back to the cylinder, draining directly to the still or draining to the base storage tank. A means shall be provided to steam strip filter cartridges within canisters prior to filter replacement.

3.4.11 Solvent level and temperature indicators. The solvent level in the cylinder shall be visible from the front of the machine and shall be calibrated in inches. High and low levels shall be highlighted. Individual temperature indi-cators shall be provided for sensing cylinder air temperature (dry sensor), drying air temperature, evaporator air temperature and solvent inlet temperature. Indi - vidual thermostatic control shall be provided for adjusting dry sensor and drying temperature settings (see 3.4.5). The drying air temperature setting shall be not greater 185°F.

3.4.12 Electrical requirements. The machine shall operate on 440 volts alternating current (Vat), 3-phase, 60 Hertz (Hz) power supply. The electrical equipment shall conform to the recommended practices for electrical installation onboard ship as specified In IEEE Standard 45. Electrical equipment shall operate on a type I electrical system as specified in DOD-STD-1399, section 300. The control panel shall be dripproof in accordance with MIL-STD-108. Pushbuttons end switches shall be arranged for safe and convenient operation to ensure operator hands are safely outside rotating cylinder area.

3.4.12.1 Electrical motors. Electric motors shall be in accordance with NEMA standard MG-1 and shall conform to the following characteristics:

Rating	-	Continuous
Insulation	-	Class B or F
Voltage	-	440 Vat, 3-phase, 60 Hz

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Enclosure	-	Dripproof
Protection	-	Thermally protected
Speed	-	
Wash/ Extract	-	Two speed, reverse not less than 3 times per 95 seconds in wash and dry mode
Pump	-	Single speed
Blower	-	Two speed

3.4.12.1.1 Wire. Motors shall be wound with wire sizes conforming to American Wire Gauge (AWG). Half size wire shall not be used.

3.4.12.1.2 Bearing. Motors shall be equipped with ball bearings, except that motors of 1/2 horsepower or less when used for horizontal applications may use sleeve bearings. Motor bearings shall be permanently lubricated type or shall have adequate and accessible means for lubrication.

3.4.12.1.3 Thermal protection. Each motor 1/8 horsepower rating or larger shall be provided with manually reset thermal overload protection incorporated es a part of the motor or installed on the controller.

3.4.12.2 Controllers. Motor controllers shall be of the magnetic type with overload protective relays. Contractors shall switch all three lines of 3-phase circuits and both sides of single phase power circuits. Motor controllers shall provide thermal overload protection for motors 1/8-horsepower and larger. Control circuits shall operate 240 volts or less. Reduce voltage control circuits shall be protected by cartridge fuses sized not greater than 200 percent of the transformer kilovoltampere (KVA) rating or 100 percent of control wire amperage rating, whichever is smaller. Start and stop pushbuttons and interlocks shall be commercially dripproof. Electrical components mounted outside the control panel shall be provided with dripproof enclosures in accordance with MIL-STD-108 or better. Solenoid valves shall be provided with epoxy (high temperature) encapsulated coils . Control boxes and electrical or air operated control valves shall be accessible from the sides (front or back) of the machine to the maximum extent practicable.

3.4.12.3 Wiring methods and practices. Wiring between different devices mounted on the machine routed outside of an enclosure shall be in rigid metal conduit or liquid tight flexible metal conduit not less than 3\8 inch electrical trade size. Fittings and junction box connections for conduit entrance shall be either threaded bushings or watertight, neoprene bushed, strain relief type. Drainage holes shall be provided at the low point of all conduit runs or junctions. Wiring shall be with moisture resistant thermoplastic insulated copper conductors rated for an operating temperature of 176°F. Conductors shall be stranded. Wiring shall be color coded and marked every 12 inches with terminal numbers. Terminal blocks shall be plainly marked to correspond with the markings on the electrical diagram. Wires run in conduit or in harness assemblies shall be marked to correspond with the electrical diagram. Solder lugs, soldered joints, and wire-nut type connections are not permitted. Electrical connections shall be made using solderless ring or pressure fit connectors to parts and terminal boards. Not more than three connections shall be made at each terminal. Conduit, fittings, and wire shall conform to and be installed in accordance with the National Electric Code. The machine shall be used on an ungrounded electrical

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system and no grounds shall be permitted on the machine wiring. Connections shall be made between motors, controllers, timers, push buttons and limit switches installed on the unit. Leads shall be extended from motors and control devices to a connection box and properly identified for connections to wiring beyond the unit. Electrical cable length shall be kept to a minimum with sufficient use of wire wraps and clamps to prevent wire insulation chaffing. The electrical controls shall be provided with a hinged l ccess door with hardware to tightly secure it closed (key or magnetic latches shall not be used). Access door shall be sized so that maintenance and repair tasks can be easily and safely accomplished (see 3.3.14).

3.4.13 Interlock and safety devices.

3.4.13.1 Functional modes. Safety and interlocking devices shall function for both clockwise and counterclockwise rotation of the cylinder and when the ship is rolling or listing at 12 degrees from the normal position.

3.4.13.2 Wash-extract interlock. The driving arrangement shall permit changing from the wash to extract cycle and vise versa with interlocking feature to ensure smooth transfer.

3.4.13.3 Brake interlock. A brake interlock switch shall be provided to prevent extraction while brake is applied.

3.4.13.4 Door interlock. A shell door interlock shall prevent the door from being opened when the cylinder is rotating and prevent rotation of the cylinder when the shell door is opened.

3.4.13.5 Guards. Exposed belts, shafts, pulleys and other moving parts shall be fully enclosed by metal guards. Gears shall be fully enclosed in gear housing. Metal guards shall be firmly supported but readily removable for maintenance.

3.4.13.6 Vibration switch. The machine shall be furnished with an adjustable vibration limiting device to disconnect service should excessive equipment damaging vibration occur.

3.4.13.7 Drain interlock. The machine drain valve shall be electrically or pneumatically interlocked to prevent the machine from entering or continuing in the extract mode when the drain valve is closed.

3.4.13.8 Still safety device. Still float, pressure relief valve and thermostat shall be as specified in 3.4.9.

3.4.13.9 Drying air safety devices. Airflow sensor, temperature limiting sensors and cylinder and blower interlocks shall be as specified in 3.4.13.9.1 through 3.4.13.9.3.

3.4.13.9.1 Airflow sensor. An airflow sensor shall be installed to interrupt the cycling of cylinder heating when the airflow through the heating coil and cylinder is not sufficient to prevent garment scorching. The airflow sensor shall

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be designed to prevent intermittent actuation when the machine is exposed to shipboard motion and vibration.

3.4.13.9.2 Temperature limit control. The drying air temperature sensor shall be installed immediately downstream of the steam heating coil to sense the temperature of the air leaving the heating coil. The sensor shall act as a primary heat sensor. The sensor shall be wired such that it will interrupt cycling of cylinder heating should the temperature exceed the specified setting. Maximum setting shall be in accordance with 3.4.11. The sensor shall automatically restore cylinder heating when the temperature falls to a safe level. The cylinder drying air sensor shall act as a secondary heat sensor. It shall interrupt cylinder heating should the temperature exceed the specified setting. The temperature setting shall be not greater than 130°F.

3.4.13.9.3 Cylinder and blower interlock. The cylinder and blower shall be individually interlocked to prevent the cycling of cylinder heating should cylinder or blower motor fail.

3.4.13.10 Emergency stop button. Emergency stop button shall be located on the front of the machine within an accessible area. Emergency stop button shall be bright red with the words “emergency stop” printed with red lettering on white background. Lettering shall be not less than 0.375 inches in height. Button shall bring machine to an immediate stop when activated. Machine shall reset with a reset button. The activation of the emergency stop button shall be identified by a red indicator light.

3.4.14 Automatic formula control unit. The unit shall be of the electro-mechanical, replaceable card type on Type I machines and a programmable microprocessor on Type II machines. The automatic formula control unit shall direct the machine through a complete operating cycle as programmed, automatically control temperatures and shall signal both audibly and visually, at the end of the cycle or at other programmed times.

3.4.14.1 Control phases. The control unit shall automatically control all phases of clutching and braking, and all motors. The unit shall automatically control the solvent level which shall be readily readjustable in the field. The control shall automatically add solvent when needed to compensate for solvent absorbed by the load. Drying air temperature and solvent temperature shall be automatically thermostatically regulated by adjusting flow of steam to recovery housing and flow of water to base storage tank respectively. Distillation and filtration shall be automatically controlled by the control unit. Automatic timing of each operation shall not proceed unless the proper solvent levels, temperatures and pressures exist.

3.4.14.2 Control mounting. The control unit shall be mounted directly to the machine. It shall permit easy viewing and accessibility and shall not interfere with the operation or maintenance of the machine. The unit mounting shall inhibit the transfer of vibration and provide a dripproof seal for the controller.

3.4.14.3 Power input and output connector. A single or separate quick locking type connector shall be used to connect the power inputs and outputs to the automatic formula controller. The configuration of the connector or connec-

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tors shall prevent incorrect wire attachment from the machine to the control unit and shall prevent accidental disconnection.

3.4.14.4 Simultaneous energizing. The control unit and machine shall be so wired that simultaneous energizing of more than one phase of a formula at a time will not occur.

3.4.14.5 Pilot light. A pilot light shall be provided to indicate the automatic timing is proceeding on Type I machines. A digital LED or LCD timer display indicating time left within the formula shall be provided on the control unit on Type II machine.

3.4.14.6 Audible and visible signal. The audible and visible signal (see 3.4.14) shall persist until canceled by the operator. Automatic timing shall not proceed until the signal is canceled. With the exception of when the machine is in the extract phase, the machine shall stop whenever the signal operates and shall restart when the signal Is canceled.

3.4.14.7 Manual control operation. Type I and Type II machines shall have a means to override automatic operation to permit total independent manual operation for complete cycle, including filtration and distillation or interruption during any automatic phase.

3.4.14.8 Microprocessor automatic control . The microprocessor automatic formula control unit shall have en operational mode for storing not less than four independent wash formulas. The formulas shall have an adequate number of separate steps to accommodate any combination of fill, wash, drain, extract, dry, deodorize, filter and/or distill. The unit shall be provided with a minimum of two preprogrammed formulas. Current wash formulas shall be obtained from Navy Exchange Command, Fleet Assist Team. Preprogrammed formulas shall be filter washed-single bath. Microprocessor shall program distillation and filtration steps during and after wash cycle. Two formula programs shall be blank. Formulas shall be retained in static random access memory, In case power loss to the machine . The back-up source shall be recharged when the power is restored to the machine. A locking mechanism shall be provided to prevent accidental reprogramming or tampering by untrained and unauthorized personnel. Programmability shall incorporate the l ase of step selection within an existing programmed formula or formula undergoing programming. The selection of limits (temperature, step time, etc) for required steps shall be programmed. If the step does not have the proper limit or limits specified, or the limits specified are unreasonable, the step shall Indicate a programing error and further programming may not commence until the error is alleviated. An end step shall indicate to the microprocessor to assign formula steps into memory. The circuitry for the micro-processor shall use bread boards which can be removed and replaced using a plug-type receptacle. The mounting of the bread boards within the microprocessor shall prevent electrical disconnect or shorting under adverse machine vibrational forces. Operation and programmability of the microprocessor shall be at an 8th grade level.

3.4.14.9 Displays, buttons, switches and audible signals. Displays, buttons, switches and audible signals on the automatic formula control units shall be as specified in 3.4.14.9.1 through 3,4.14.9.15.

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3.4.14.9.1 General criteria. Controls shall be selected and integrated into the machine in order to meet the applicable human engineering requirements specified in ASTM F 1166.

3.4.14.9.2 Distribution of work load. Controls shall be selected and distributed so that none of the operator limbs will be overburdened.

3.4.14.9.3 Stops. Stops shall be provided at the beginning and end of the range of control positions if the control is not required to be operated beyond the indicated end positions or specified limits.

3.4.14.9.4 Consistency of movement. Direction of control movement shall be consistent with the related movement of an associated display or equipment component. Movement of a control forward, clockwise, to the right, or up, or pressing a control shall turn the equipment or component on or cause the quantity to increase.

3.4.14.9.5 Operator-control orientation. Controls shall be oriented with respect to the operator.

3.4.14.9.6 Grouping. Controls that function in sequential operation necessary to a particular task or that operate together shall be grouped together along with their associated displays. When several steps of a sequence are selected by one control, the steps shall be arranged by order of occurrence to minimize control movements and prevent cycling through unnecessary steps. Cycling through the control's ON or OFF position shall be avoided.

3.4.14.9.7 Sequential operation. Where sequential operations follow a fixed pattern, controls shall be arranged to facilitate operation (for example, in a pattern left-to-right and top-to-bottom, like a printed page).

3.4.14.9.8 Location of primary controls. The most important and frequently used controls shall have the most favorable position with respect to case of reaching and grasping (particularly rotary controls and those requiring fine settings). When a transilluminated indicator is associated with a control, the indicator light shall be so located that it may be associated with the control without error and shall be visible to the operator during control operation.

3.4.14.9.9 Spacing. Minimum spacing between controls shall be as specified in ASTM F 1166. Spacing between a control and any adjacent obstruction shall be as shown on the figures in ASTM F 1166. The spacing between adjacent edges of simple, round indicator light fixtures shall permit unambiguous labeling, signal interpretation, and convenient bulb removal.

3.4.14.9.10 Relation to display. When color-coding must be used to relate a control to its corresponding display, the same color shall be used for both the control and the display.

3.4.14.9.11 Control panel contrast. The color of the control shall be black or gray and shall provide contrast between the panel background and the control.

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3.4.14.9.12 Labeling of controls. Controls and displays shall be appropriately and clearly labeled with plates that contain the basic information needed for proper identification, utilization, actuation, or manipulation of the element. By using simple words and phrases the labels shall convey verbal meaning in the most direct manner. Bach control and display shall b. labeled according to function, and the following criteria shall apply.

a. Similar names for different controls and displays shall be avoided.

b. Instruments shall be labeled in terms of what is being measured or controlled taking into account the user and purpose.

c. Control labeling shall indicate the functional result of control movement (for example, ON or OFF). Such information shall be visible during normal operation of the control.

d. Labels shall be placed above the controls ‘and displays they describe and shall be oriented horizontally so that they may be read from left to right. When the panel is above eye level and if label visibility will be enhanced, labels may be located below the controls.

e. The units of measurement (for example, degrees Fahrenheit (°F) or minutes) shall be labeled on the panel.

f. Lettering shall be black on a light background.

3.4.14.9.13 Prevention of accidental actuation. Controls shall be constructed and located so that they are not susceptible to being moved accidentally (for example, controls shall be provided with resistance such as viscous or coulomb friction, spring-loading and so forth).

3.4.14.9.14 Coding. Simple indicator lights shall be coded to conform to the requirements specified in ASTM F 1166.

3.4.14.9.15 Intensity of audio signals. Audio warning signals shall not be so intense that discomfort or ringing in the ears is caused.

3.4.14.10 Microprocessor and control panel layout. As a minimum the following displays, buttons and switches shall be provided:

Power - A display light and switch/button shall be provided to turn on and off the electrical power to the microprocessor.

Mode - A switch to allow operator to set controller to automatic or manual mode.

Formula - Allows operator to select a formula from memory. A indicator display shall be provided to identify the wash formula under selection and operation.

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Step - A indicator display shall be provided to identify the stop number of the formula. A switch shall be provided to advance through formula one step at a time.

Timers - Digital displays shall be provided to indicate the total time remaining in the present step under operation and complete wash formula. The timers shall descend when the formula is in operation and shall have a time increment of not greater than one minute. The timers shall also be used to program the time for each required step within the wash formula.

Solvent level - Display shall be provided to indicate solvent height selected (low, medium or high) for selected wash formula and shall also be used to program fill height within wash formulas.

Signal - A push button shall be provided to cancel the signal. A signal display Shall be used to program signal operation within a formula.

Start/Resume - A push button shall be provided to start a selected formula which has been stopped.

stop - A push button shall be provided to stop a selected formula in progress.

Reset - A push button shall be provided to reset a stopped formula to it’s beginning to be restarted or to allow selection of another formula.

Extract - A display shall be provided to indicate the extract cycle of a formula.

Temperatures - A digital display shall be provided to indicate drying air, dry sensor, evaporator air and solvent inlet temperatures. Thermo-static controls shall be provided to regulate drying air and dry sensor temperatures.

Pressure - Digital displays shall be provided to indicate pressure of cooling water to base tank, still condenser and refrigeration condenser. Pressure gages shall be interlocked with audible and visible alarms which will activate If pressure exceeds specified setting.

3.4.14.10.1 Manual switches and indicatorsindicators. Displays, buttons and switches specified in 3.4.14.10 shall be used to the maximum extent practical to meet the manual operating requirement specified in 3.4.14.7. Additional switches and corresponding indicator lights shall be provided as required.

3.4.14.10.2 Diagnostic displays. Individual indicator lights shall be provided which correspond to each valve, motor or solenoid to indicate operational status of component. Minimum number of indicators shall be provided to allow operator to accomplish full diagnostic troubleshooting,

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3.4.14.11 Electro-mechanical automatic control. The electro-mechanical automatic control units shall have operational functions and capabilities similar to microprocessor automatic control unit in both automatic and manual modes of operation.

3.5 Performance characteristics. Performance characteristics shall be as specified in 3.5.1 through 3.5.9.

3.5.1 Automatic controller. The control unit shall meet the following shipboard requirements (see 4.4.2.1).

Test	Performance requirement
Low temperature	To determine if controller can be stored and operated under the conditions specified in 4.4.2.1.1 without experiencing physical damage or deterioration in performance.
High temperature	To determine if controller can be stored and operated under the conditions specified in 4.4.2.1.2 without experiencing physical damage or deterioration in performance.
Humidity	To determine controller's ability to perform in hot and humid atmosphere (see 4.4.2.1.3) without physical damage or deterioration in performance.
Enclosure	Failure of control unit to operate or accumulation of water within the enclosure shall be cause for rejection (see 4.4.2.1.11).
Steady state voltage and frequency	Control unit shall operate satisfactorily after each test condition (see 4.4.2.1.4).
Transient voltage and frequency	Control unit shall return to normal operation without deterioration in performance after transients (see 4.4.2.1.5 and 4.4.2.1.6).
Spike voltage	Control unit shall return to normal operation without deterioration in performance after spikes on each phase (see 4.4.2.1.7).
Power interruption	Control unit shall return to normal operation without deterioration in performance when power is restored (see 4.4.2.1.8).
Insulation resistance	Control unit shall have an insulation resistance of not less than 10 megohms (see 4.4.2.1.9).
Leakage current	Control unit under normal operation shall have a leakage current not greater than 5 milliamperes (ma) (see 4.4.2.1.10).

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3.5.2 Inclined operation. The machine shall function in accordance with the requirements of this specification, without reduction in performance when inclined at 12 degrees (see 4.4.2.2).

3.5.3 Vibration. The machine, with the automatic formula control unit, shall meet the applicable requirements of MIL-STD-167-1 from 4 to 33 hertz (Hz) for type I vibration. Equipment orientation for vibration test requirements specified in MIL-STD-167-1 may be based on the assumption that the machine will be installed in the ship with the axis of the cylinder rotation parallel to the centerline of the ship (see 4.4.2.3).

3.5.4 Efficiency. The machine, under full load operation, shall comply with the Environmental Protection Agency emission standards and Occupational Safety and Health Administration exposure limits for use of perchloroethylene (see 6.2).

3.5.5 Airborne noise. The machine shall be not greater than the octave band sound levels specified in MIL-STD-740-1 for grade D equipment when tested under full load capacity conditions in both wash and extract modes of operation (see 4.4.2.4).

3.5.6 Electromagnetic comparability. The machine shall be tested to the applicable electromagnetic interference requirements of MIL-STD-461, class A4 for the collection of CE01, CE03 and RE02 data (see 4.4.2.5).

3.5.7 Electric motors. Electric motors shall demonstrate their ability to perform and shall be not greater than their maximum rated draw (see 4.4.2.6).

3.5.8 Opration. Each machine shell perform all its operations satisfactorily (see 4.4.2.8). Operation of the machine shall not burden an operator within the fifth to ninety-fifth athropedic dimensions by requiring excessive reaching for controls and visually identifying settings or gauge reading.

3.5.9 Saturated steam. Pipes, fittings, coils and chambers using steam shall be suitable for 100 lb/sq in saturated steam and show no leakage when tested to a pressure of 150lb/sq in. (see 4.4.2.7).

3.6 Identification plate. Identification plates shall be made for the machine and its major components; for example, electric motor, refrigeration components and so forth. The plates shall be made of corrosion-resistant steel, aluminum alloy or brass. Characters shall be not less than 3/16 inch in height. Characters shall be stamped on the plate. Stamping shall not be less than 0.003 inch deep. The thickness of the plate shall be not less than 0.03 inch. Plate information shall remain permanent and legible for the life of the equipment.

3.6.1 Mountings. The identification plate shall be securely affixed to each machine in a conspicuous place.

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3.6.2 Identification plate date. Data marked on the identification plate shall include the following:

a. Manufacturer's type or model number, serial number, contract or order number, and date of manufacturer.

b. Salient design features including size, voltage, current, refrigerant and capacity in pounds.

c. Name of machine and stock number (the stock number will be provided to the contractor).

d. Component Identification Number (CID). The CID, followed by a nine digit number that will be furnished upon receipt of provisioning technical documentation. No equipment shall be shipped without the CID number for the machine, and its associated components shall also be recorded on the packing list.

3.7 Workmanship. The construction, fabrication, and finish shall be suitable for the purpose intended and shall be of the highest quality commensurate with the intended use. Tolerances shall be in accordance with good commercial standards. In general, workmanship shall be equivalent to the best in the industry and shall conform in detail to the following:

a. WeldS shall be cleaned and ground to present a uniform and smooth finish.

b. Methods of fastening components in place shall provide the strength and rigidity necessary to resist local stresses, internally or externally induced vibration, and impact forces. Fasteners shall provide the degree of demountability required for maintenance and repair that is not provided by removable or hinged accesses.

c. Rotating parts shall be aligned and balanced in order to prevent vibration and eccentric motion. With regard to cylinder rotation this requirement shall apply to both empty and loaded conditions. Bearings, gears and shafts shall be properly constructed, aligned and fitted for durable operation.

d. Surfaces and the cylinder interior shall be free of burrs and sharp edges.

e. Chains and belt drive adjustment mechanisms shall be readily accessible and efffective as specified in 3.3.14.

f. Access plates and doors shall be provided as necessary for proper maintenance, cleaning, and lubrication and shall be a suitable size and properly located.

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g. Safety guards shall be adequate and suitably located.

h. lubrication fittings shall be provided where necessary and adequate for the purpose.

4. QUALITY ASSURANCE PROVISIONS

4-1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program, The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable paragraphs in this specification.

4.4 First article inspection. First article inspection shall consist of the examinations shown in table I and tests specified in 4.4.2.

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TABLE I. Classification of defects.

EXAMINATION	DEFECT (REQUIREMENT)	CLASSIFICATI	
		MAJOR	MIN
Materials	Construction materials not as specified (see 3.2 through 3.2.13).	X	
Design, general	Configuration not as specified (see 3.3 through 3.3.14).	X	
Construction/ Mechanical	Shell not as specified (see 3.4.1 through 3.4.1.3).	X	
	Cylinder not as specified (see 3.4.2 through 3.4.2.1.1).	X	
	Cylinder bearings are not readily replaceable (see 3.4.2.1.3)	X	
	V-belts are not readily adjustable (see 3.4.2.1.4).	X	
	Brake not as specified (see 3.4.3).	X	
	Cooling water piping and connections not as specified (see 3.4.4).	X	
	Steam heating units not as specified (see 3.4.5).	X	
	Blower not as specified (see 3.4.6)	X	
	Refrigeration system is not dehy-rated and charged (see 3.4.7.1.1).	X	

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TABLE I. Classification of defects - Cont'd.

EXAMINATION	DEFECT (REQUIREMENT)	CLASSIFICATI	
		MAJOR	MIN
Construction/ Mechanical	Refrigeration condenser is not designed for use with inlet water at 85°F (see 3.4.7.1.2)	X	
	Evaporator cooling coil not as specified (see 3.4.7.1.3).	X	
	Lint filter not as specified (see 3.4.7.2)	X	
	Base storage tank in-tank cooling coil is not provided with an automatic thermostatic valve (see 3.4.8).	X	
	Distillation rate of still not as specified (see 3.4.9).	X	
	Means to steam strip filter cartridges not provided (see 3.4.10).	X	
	Temperature indicators and thermostatic controls not as specified (see 3.4.11).	X	
Construction/ electrical	Power supply not as specified (see 3.4.12).	X	
	Motors not as specified (see 3.4.12.1).	X	
	Control panel and electrical components not dripproof as specified (see 3.4.12 and 3.4.12.2)	X	
	Wiring conduit not as specified (see 3.4.12.3)	X	
	Terminal blocks not marked to correspond with electrical diagram and wires not marked to correspond with electrical diagram (see 3.4.12.3).	X	

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TABLE I. Classification of defects - Cont'd.

EXAMINATION	DEFECT (REQUIREMENT)	CLASSIFICATI	
		MAJOR	MIN
Interlock and safety devices	Door interlock not as specified (see 3.4.13.4)	X	
	Wash-extract interlock not as specified (see 3.4.13.2)	X	
	Brake interlock not as specified (see 3.4.13.3)	X	
	Belts, shafts, pulleys are not fully enclosed. Guards are not readily removable (see 3.4.13.5)	X	
	Vibration switch not as specified (see 3.4.13.6)	X	
	Drain interlock not as specified (see 3.4.13.7).	X	
	Still safety devices not as specified (see 3.4.13.8)	X	
	Drying air safety devices not as specified (see 3.4.13.9 through 3.4.13.9.3)	X	
	Emergency stop button not as specified (see 3.4.13.10).	X	
Automatic formula control unit	Automatic formula control unit not as specified (see 3.4.14).	X	
	Control phase not as specified (see 3.4.14.1).	X	
	Control mounting not as specified (see 3.4.14.2).	X	
	Power input and output connector not as specified (see 3.4.14.3).	X	
	Simultaneous energizing wiring not as specified (see 3.4.14.4).	X	
	Pilot light not as specified (see 3.4.14.5).	X	

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TABLE I. Classification of defects - Cont'd.

EXAMINATION	DEFECT (REQUIREMENT)	CLASSIFICATI	
		MAJOR	MIN
Automatic formula control unit	Audible and visible signals not as specified (see 3.4.14.6).	X	
	Manual control operation not as specified (see 3.4.14.7).	X	
	Microprocessor automatic control not as specified (see 3.4.14.8).	X	
	Microprocessor controller layout not as specified (see 3.4.14.10 through 3.4.14.10.2)	X	
	Electro-mechanical automatic control not as specified (see 3.4.4.11).	X	
	Displays, buttons, switches, and audible signals not as specified (see 3.4.14.9 thru 3.4.14.9.15).	X	
Identification	Identification and information plate not of specified material (see 3.6)		X
	Markings illegible (see 3.6).		X
	Identification plate mounting not as specified (see 3.6.1).		X
	Identification plate data not as specified (see 3.6.2).		X
Workmanship	Welds not uniform and smooth (see 3.7 a).		X
	Method of fastening components not as specified (see 3.7 b).		X
	Rotating parts are not aligned and balanced as specified (see 3.7 c).		X
	Cylinder interior not free from burrs or sharp edges (see 3.7 d).		X
	Chains and belt drive adjustment mechanisms not readily accessible and effective (see 3.7 e).		X

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TABLE I. Classification of defects - Cont'd.

EXAMINATION	DEFECT (REQUIREMENT)	CLASSIFICATI	
		MAJOR	MIN
Workmanship	Access plates and doors not as specified (see 3.7 f).		X
	Safety guards are not as specified (see 3.7 g).		X
	Lubrication fittings are not as specified (see 3.7 h).		X

4.4.1 First article sample. Prior to the start of production and wit 30 days after the date of award of the contact or order, one machine of each design and size shall be made ready for approval of the contracting officer his authorized representative for the tests and examinations specified in ta and 4.4.2. Disposition instructions for first article sample shall be obtai from the contracting officer's technical representative (see 6.2).

4.4.2 First article inspection test. First article inspection tests shall consist of tests as specified in 4.4.2.1 through 4.4.2.9.

4.4.2.1 Test for automatic formula control unit. The automatic formu control unit shall meet the requirements specified in 3.5.1 when tested in accordance with 4.4.2.1.1 through 4.4.2.1.11.

4.4.2.1.1 Low temperature tests. Unit shall be tested in accordance method 502, procedure I and II, of MIL-STD-810. Unit shall be tested at a co stant storage temperature of minus 4°F for 24 hours, and a constant operating temperature of 32°F for 24 hours (see 6.3).

4.4.2.1.2 High temperature test. Unit shall be tested in accordance method 501, procedures I and II, of MIL-STD-810. Unit shall be tested at a c stant storage temperature of 160°F for 24 hours, and a constant operating temperature of 122°F for 24 hours (see 6.3).

4.4.2.1.3 Humidity. Unit shall be tested in accordance with method 50 procedure I, of MIL-STD-810. Unit shall be tested at relative humidity of 95 percent or higher at temperature range of 86 to 122°F (see 6.3).

4.4.2.1.4 Steady state voltage and frequency. The equipment shall be operated for not less than 15 minutes in each of the conditions A thru E of t II. Performance readings shall be measured for each condition. Failure in a performance parameter shall be cause for rejection.

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TABLE II. Steady state voltage and frequency.

Condition	Rated Voltage (+/- 10%)			Frequency (60 Hz) (+/- 5%)		
	Lower limit	Normal	Upper limit	Lower limit	Normal	Upper limit
A (reference condition)		X			X	
B	X			X		
C			X	X		
D			X			X
E	X					X

4.4.2.1.5 Transient voltage.

4.4.2.1.5.1 Upper limit. With the equipment operating in the upper li of steady state voltage, a transient voltage of plus 18 percent of normal vol recovering to the steady state voltage within 2 s shall be superimposed. The equipment shall operate normally following the transient.

4.4.2.1.5.2 Lower limit. With the equipment operating on the lower li of steady state voltage, a transient voltage of minus 18 percent of normal vo recovering to the steady state voltage within 2 s shall be superimposed. The equipment shall operate normally following the transient.

4.4.2.1.5.3 Transient (700V). In addition to the tests of 4.4.2.1.5.1 4.4.2.1.5.2, equipment containing transistors or semi-conductor devices shall subjected to an input supply line transient of 700V amplitude and 2 ms durati while the equipment is operating at normal supply line voltage and frequency equipment shall operate normally immediately following the transient.

4.4.2 .1.6 Transient frequency.

4.4.2.1.6.1 Upper limit. With the equipment operating at 5 percent ab the normal frequency, the frequency shall be increased by an additional 3 per recovering to the steady state frequency (+/- 5 percent of normal) within 2 s The equipment shall operate normally following the transient.

4.4.2.1.6.2 Lower limit. With the equipment operating at 5 percent be the normal frequency, the frequency shall be decreased by an additional 3 per recovering to the steady state frequency (+/- 5 percent of normal) within 2 s The equipment shall operate normally following the transient,

4.4.2.1.7 Spike voltage test. In addition to the tests of 4.4.2.1.6.1 4.4.2.1.6.2 the equipment shall be subjected to an input supply line voltage

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of 2500V position peak amplitude; the wave shape shall correspond with that figure for the “spike voltage (short time transient) wave shape”, section 30 DOD-STD-1399. This spike shall be impressed at normal supply line voltage a frequency while the equipment is in operation. The equipment shall operate normally immediately following the test.

4.4.2.18 Power interruption. With the equipment operating within th steady state tolerances of voltage and frequency, the external power supply be suddenly interrupted and after an interval between 3 and 4 s, the power s within the steady state tolerances. shall be reappliad. After the equipment been operated long enough to detect any major performance degradation and to include any recycling time, the power shall be interrupted for an interval o to 30 s. The cycle, consisting of three such interruptions, shall be repeat a total of four times. During and as a result of these tests, no damage to equipment shall be incurred, and any noted effects of power interruption or reapplication shall fall within the performance tolerances of the individual equipment specification. Where an equipment has more than one significant operating mode, the power interruption tests shall be performed in each of t modes.

4.4.2.1.9 Insulation resistance. The insulation resistance of electr circuits shall be not less than 10 megohms at 500Vdc at a temperature of 23 degrees +/- 1.4 degree C and at a relative humidity of 50 +/- 5 percent. Al synchros shall be disconnected during this test.

4.4.2.1.10 Leakage current. The leakage current between the power li equipment enclosure shall be measured through a 1500 ohm resistive load at n power line voltage. Leakage current of the equipment shall be not greater t 5mA to ground.

4.4.2.1.11 Enclosure. Unit shall be tested in accordance with MIL-STD "drip test". The angle of inclination shall be 45 degrees.

4.4.2.2 Inclined operation. The machine shall be inclined at an angl 12 degrees to each side of two vertical planes at right angles to each other shall operate normally through a complete wash cycle, including filtration a distillation, with capacity load in each of the four positions (see 3.5.2).

4.4.2.3 Vibration. The machine shall be tested in accordance with the requirements specified in MIL-STD-167-1 type I vibration. The machine shall operationally run during and after the test (see 3.5.3 and 6.3).

4.4.2.4 Airborne noise. The machine shall be tested for airborne nois produced during operation in accordance with the procedures specified in MIL-STD-740-1. Measurements shall be for sound pressure level determination The machine shall be tested under full load conditions in both wash and extra modes of operation. The machine shall meet the requirements of 3.5.5 (see 6.

4.4.2.5 Electromagnetic interference. MIL-STD-461, class 4A, test CE0 CE03 and RE02 shall be performed on the machine, Graphical limiting emission surface ships shall be as specified in MIL-STD-461. Test setups shall be as specified in MIL-STD-462 (see 3.5.6 and 6.3).

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4.4.2.6 Power input. After accelerating to full speed under maximum load conditions, the motor(s) shall not draw in excess of the rated current (see 3.5.7).

4.4.2.7 Operation The machine shall be operated for three complete automatic cycles and three complete manual cycles, including filtration and distillation. During this test it shall be verified that all valve, controls, switches, interlocks, gauges, temperatures, pumps and motors are functioning properly and that there are no solvent, water, steam or refrigerant leaks. Satisfactory operation of the refrigeration recovery circuit, filtration system and still shall be verified (see 3.5.8).

4.4.2.8 Saturated steam. Steam coils and piping systems shall be hydrostatically tested to a pressure of 150lb/sq in for not less than 30 minutes. In both cases no leaks in either system shall be permitted (see 3.5.9).

4.4.2.9 Limiting devices. Drying air safety devices shall be tested by placing the machine into the following operational condition until automatic shutdown occurs:

a. Restricted airflow (see 3.4.13.9.1)
b. Blower failure (see 3.4.13.9.3)
c. Failure of cylinder rotation (see 3.4.13.9.3)

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the examinations of 4.5.2 and the tests of 4.5.3. (see 6.3 and 6.6).

4.5.1 Lot. For the purpose of quality conformance inspection and test sampling, a lot is defined as all the machines of one type, produced in one facility, using the same material and production processes, and being offered for delivery at one time.

4.5.2 Sampling for visual and mechanical examination. A random sample of machines shall be selected in accordance with Table III and examined in accordance with Table I. If one or more major defects are found in any sample, the entire lot shall be rejected. The contractor has the option of screening the lot 100% for the defective characteristic(s) or providing a new lot which shall be inspected in accordance with the sampling plan contained herein. The contractor shall maintain for a period of three years after contract completion all records of inspections, tests, and any resulting rejections,

TABLE III. Sampling for visual and mechanical examination and operational test.

Lot size	Sample size
2 to 8	All
9 to 90	8
91 to 150	12
151 to 280	19
281 to 500	21
501 to 1200	27

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4.5.3 Sampling for operation test. A random sample of machines shall be selected in accordance with Table III and tested in accordance with paragraph 4.5.3.1. If any machines fails any of the tests specified in paragraph 4.5.3.1, the entire lot shall be rejected. The contractor has the option of screening the lot 100% for the defective characteristic(s) or providing a new lot which shall be inspected in accordance with the sampling plan contained herein. The contractor shall maintain for a period of three years after contract completion all records of inspection, test, and any resulting rejection.

4.5.3.1 Operational testing. Each machine from the random sample shall be connected to utility services and operationally tested without garments or laundry. The following items and operations shall be checked for compliance to the requirements specified:

a. Vapor containment (see 3.3.9)
b. Brake (see 3.4.3)
c. Blower (see 3.4.6)
d. Wash-extract interlock (see 3.4.13.2)
e. Door interlock (see 3.4.13.4)
f. Vibration switch (see 3.4.13.6)
g. Emergency stop button (see 3.4.13.10)
h. Still pressure relief valve (see 3.4.9)
i. Drying air safety devices (see 3.4.13.9)
Restricted airflow (see 3.4.13.9.1)
Drying thermostats (see 3.4.13.9.2)
Cylinder and blower interlock (see 3.4.13.9.3)
Hydrostatic testing (see 3.4.5)

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition, For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.8).

5.1 Preservation, packing and marking. The machine shall be Preserved, level A or commercial, packed level A, B, or commercial, as specified (see 6.2), and marked for each level of protection in accordance with MIL-L-3153. The machine shall be shipped fully assembled, unless disassembly Is necessary to protect certain components during transit, or disassembly results in more than a 25 percent savings in transportation costs to the Government or to the contractor. In addition, for Navy acquisitions, the following applies:

(a) Navy fire-retardant requirements.

(1) Lumber and plywood. When specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category I - general use.

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Level C - Type I - non-weather resistant.
Category I - general use.

(2) Fiberboard. Unless otherwise specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, and cleated fiberboard boxes including Interior packaging forms shall meet the flame spread index and specified optic density requirements of PPP-F-320 and amendments thereto.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Commercial laundry dry cleaning machines acquired under this specification are intended for shipboard use. Therefore , commercial machines offered to the Government may require alteration to meet specification requirements, for example, saltwater input, steam injection, proper materials, wiring, electromagnetic emission requirements, safety features.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Type, class and style of machine required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article inspection is required (see 3.1).
e. Disposition of first article (see 4.4.1).
f. When OSHA and EPA compliance certification is required (see 3.5.4).
g. Level of presevation, level of packing, and packaging options required (see 5.1).
h. When fire retardant treated lumber and plywood is required (see 5.1.a).
i. When fire retardant fiberboard is not required (see 5.1.a).
j. When technical manual is required (see 6.4).

6.3 Consideration of data requirement. The following data requirements should be considered when this specification is applied on a contract. The Applicable Data Item Descriptions (DID's) should be reviewed In conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title
3.5.4	DI-MISC-80678	Certification/Data

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4.4.2.5	DI-EMCS-80200	Electromagnetic Interference Test Report
4.4.2.3	UDI-T-23762	Report, Vibration Testing
4.4.2.1.1, 4.4.2.1.2 and 4.4.2.1.3	DI-ENVR-80863	Environmental Test Report
4.4.2.4	DI-HFAC-80272	Equipment Airborne Sound Measurements Test Report
6.6	DI-V-7002	Provisioning Parts List

The above DID's were those cleared as of the data of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit of each configuration and size. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first article. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or teat, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.6. Provisioning. Provisioning Technical Document (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.6.1 Spare or repair parts. When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 National stock number. The national stock number shown in the acquisition document is for reference only. Contractors will be advised of the applicable national stock number to be used on identification plates.

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6.8 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.9 Subject term (key word) listing.

Microprocessor
Perchlorethylene
Vapor containment
Solvent recovery
Distillation
Filtration
Compressor
Evaporator

6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 3510-N337

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-D-18163B(SH)	1994 February 23

3. DOCUMENT TITLE DRY CLEANING UNIT ASSEMBLY, FULLY AUTOMATIC, SYNTHETIC SOLVENT (PERCHLORETHYLENE) TYPE, (20-35 POUNDS CAPACITY) (SHIPBOARD)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
TECHNICAL ENGINEER: Mr. Joseph Bowen, NAVSSES 072D	(1) Commercial (215) 897-7925	(2) AUTOVON 443-7925
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command NAVSEA 03R42, 2531 Jefferson Davis Hwy Arlington, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous editions are obsolete 198.2

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